BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1
Duration: 30 minutes

Semester: Summer2025
Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1. MIPS for computer A is 6×10^9 .

6 ins.

Add ....
Sub ....
Add ....
Addi ....
Add ....
Sub ....

Find the execution time of the given instruction set in Computer A? [3]
Answer:

$6 \times 10^9 \times 10^6$ ins — 1s

1 " — $\frac{1}{6 \times 10^{15}}$ s

6 " — $\frac{1 \times 6}{6 \times 10^{15}}$ s

$= 1 \times 10^{-15}$ s

OR

$MIPS = \frac{\text{Ins. Count}}{\text{Exe. time} \times 10^6}$

$\Rightarrow \text{Exe. time} = \frac{\text{Ins. count}}{MIPS \times 10^6}$

$= \frac{6}{6 \times 10^9 \times 10^6}$

$= 1 \times 10^{-15}$ s

2. Suppose you have visited your friend's startup, which is a server management company. You observe that their server machines are highly efficient and respond to client requests with very low latency. However, you also learn that a recent electrical failure damaged the power supply of one of their main servers, resulting in nearly two hours of downtime.
As a computer architect, which of the 7 Great Ideas would you recommend your friend implement to ensure no such event occurs in the future? [2]
Answer:

Dependability via redundancy.

3. Consider the execution of Program A, which consists solely of add instructions, on two different computer systems: a reference computer and your computer. On the reference computer, running Program A requires a total of 2×10^9 instructions. The reference computer operates with a clock rate of 3 GHz and has a CPI of 3.

On your computer, the total number of clock cycles required to execute Program A is 7×10^9. Your computer has a clock duration of 2 seconds, and it is observed that the execution time to run Program A on your computer is 5 seconds.

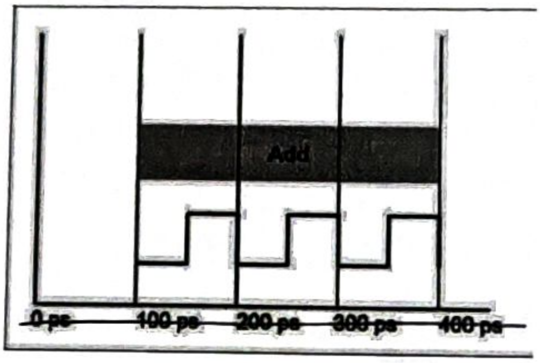

Figure - 1 : Duration of one ADD instruction in your Computer ~~A~~

a) Find the reference time. [3]

b) What is the CPI of Add instruction for your Computer ~~A~~? [2]

c) Find the clock rate of your computer. [2]

d) Find the spec ratio. [2]

e) Comment about the performance of your computer based on the spec ratio. [1]

Answer:

a) Reference time is the execution time of the reference computer.

For ref. com, Instruction count = 2×10^9

Clock Rate = 3GHz = 3×10^9 Hz

CPI = 3

$$\therefore \text{Ref. time} = \frac{IC \times CPI}{C.R}$$

$$= \frac{2 \times 10^9 \times 3}{3 \times 10^9}$$

= 2s (Ans)

b) CPI = 3

c) $\text{Clock Rate} = \dfrac{1}{\text{Clock Duration}}$

$= \dfrac{1}{2} = 0.5\ Hz$

OR

$\text{CPU time} = \dfrac{\text{Clock Cycle}}{\text{Rate}}$

$\Rightarrow 5 = \dfrac{7 \times 10^9}{\text{Rate}}$

$\Rightarrow \text{Rate} = \dfrac{7 \times 10^9}{5}$

$= 1.4 \times 10^9\ Hz$

Well the answers should have been same but as I have set the CPU time randomly to 5s, Hence the different answers.
If you used a correct approach, you will get full marks.

d) $\text{Spee Ratio} = \dfrac{\text{Ref. time}}{\text{Ex. time}}$

$= \dfrac{2}{5} = 0.4$

e) $0.4 < 1$; So, my computer is slower.

Or, Prog. A runs slower in my computer than the ref. computer.